Slopes in steep terrain are often converted into terraces. Terraces are structures that are built to divide a slope into short and gently sloping segments, as a measure to reduce the slope steepness. Terraces change the topography of slopes and thus hydrological pathways, for example by decreasing the runoff. Moreover, the surface roughness is altered, decreasing overland flow connectivity. These changes will reduce the runoff and improve water conservation, which will lead to a higher soil moisture and water holding capacity (Wei et al., 2016). Therefore, the mentioned factors will reduce (severe) soil erosion (e.g. sheet and rill erosion, gully formation) and benefit soil conservation (FAO, 2000). Worldwide, these structures are used to decrease runoff and combating soil erosion (Wei et al., 2016).

Within the tropics, however, very little is known about the impact of terracing on the occurrence of landslides. In South-East Asia, Sidle et al. (2006) argued that terraces allow for greater water infiltration and thus create conditions favourable for landslides. Turkelboom et al. (2008) observed an elevated landslide frequency on terraced slopes in northern Thailand. A similar perception is held by farmers in Eastern Uganda (Kitutu et al., 2011). It is to be expected that the impact of terraces on landslides is, to some extent, dependent on the terrace typology and management. However, this hypothesis still has to be tested in the tropics. The impact of terrace typology on land degradation processes has already been demonstrated in Rwanda where labour-intensive bench (radical) terraces appear to be more effective at controlling soil erosion compared to progressive terraces (i.e. terraces gradually made by local farmers – aka Fanya-juu terraces ), yet landslides were not investigated in these studies (Fashaho et al., 2020; Rutebuka et al., 2021).

One of the key issues associated with terracing is their abandonment (Wei et al., 2016). Such abandonments generally equal to a total lack of maintenance, which in the long run can accelerate the formation of existed rills, interrills, gullies, gravitational erosion, piping and landslides on marginal slopes. Without adequate maintenance, various natural or other human-generated forces will gradually damage the structure and strength of terrace walls and risers, leading to a complete terrace failure. For example, Agnoletti et al. (2019) showed that abandoned terraces are particularly sensitive to slope failures, yet their study is situated in temperate regions.

Another agricultural activity that can lead to slope destabilization is irrigation, which can help increase water recharge in the upper soil layers. This process has been well documented in Indonesia, where the liquefaction of unconsolidated materials associated with the 2018 Palu earthquake was accentuated in irrigated areas, triggering translational landslides and lateral extension on relatively gentle slopes of less than a few degrees (Watkinson and Hall, 2019).